

Barzolvolimab Treatment Improves Urticaria Activity in Patients with Chronic Spontaneous Urticaria Regardless of Baseline IgE Levels

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Background

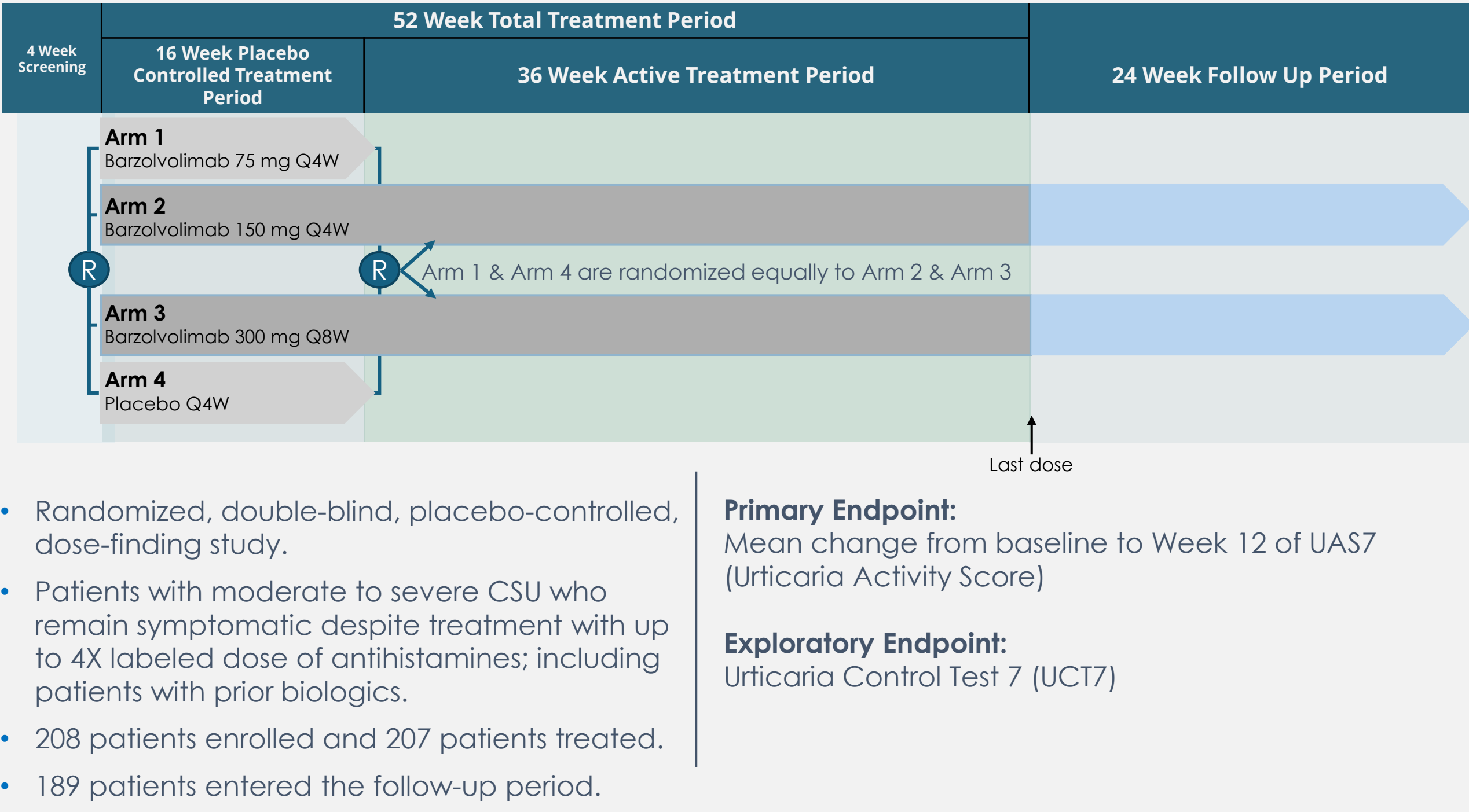
- Mast cells (MCs) are key effector cells in chronic spontaneous urticaria (CSU).
- Barzolvolimab, a humanized monoclonal antibody that inhibits KIT activation by SCF, demonstrated rapid and durable depletion of skin MCs¹.
- We previously reported that barzolvolimab treatment resulted in statistically significant and clinically meaningful improvement in weekly Urticaria Activity Score (UAS7) at 12 weeks, with deepening of response over 52 weeks in antihistamine-refractory CSU patients (NCT05368285), including patients who received prior omalizumab.
- Low circulating IgE levels have been associated with autoimmune (Type 2b) disease and refractoriness to anti-IgE therapies^{2,3,4}.
- Here we report UAS7 and Urticaria Control Test over 7 days (UCT7) results in relation to baseline total IgE level at Weeks 12 and 52 for the two higher barzolvolimab doses.

References

¹ Terhorst-Molawi et al. Allergy 2023. 78(5):1269-1279; ² Kolkhir et al. Clin Rev All Clin Imm 2022. 149(6):1819-1831; ³ Schoepke et al. Allergy 2019. 74(12):2427-2436. ; ⁴ Ertas et al. 2017 Allergy. 73(3):705-712

CSU Phase 2 Study Design and Key Eligibility

Randomized Placebo-controlled Phase 2 Study



Methods and Baseline Characteristics

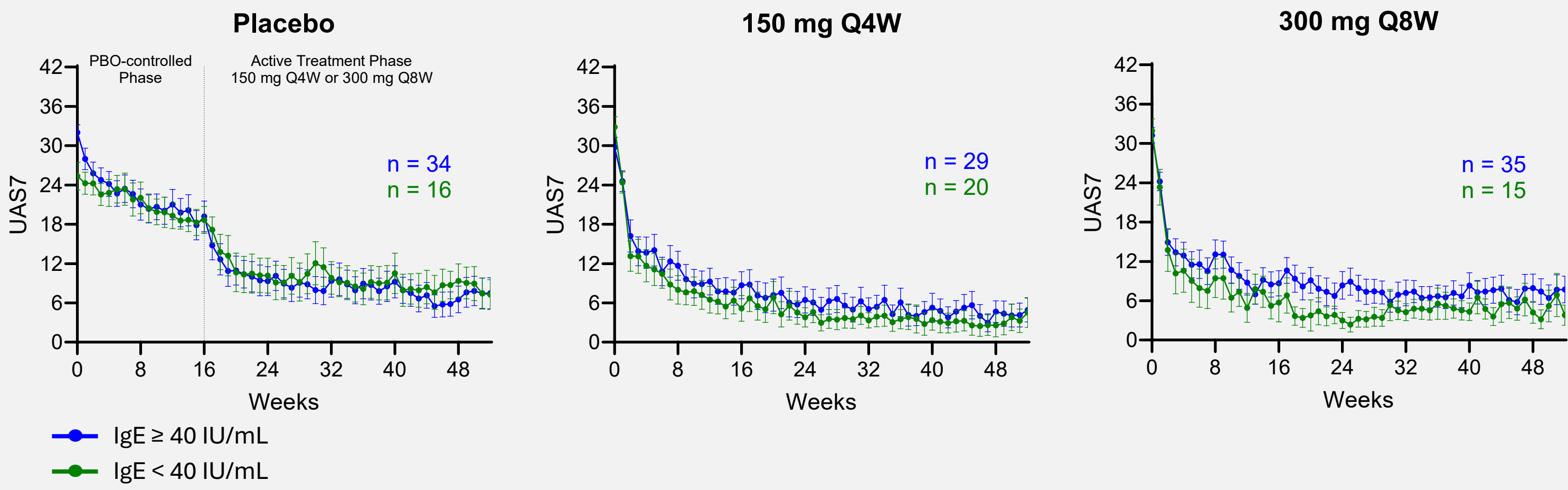
- Patients receiving placebo, or barzolvolimab at 150 mg Q4W or 300 mg Q8W were subgrouped according to baseline IgE levels as previously defined^{2,3}
 - IgE normal/high ≥ 40 IU/mL
 - IgE low < 40 IU/mL
- Analysis was conducted in patients with observable data Weeks 12 and 52. Analysis included patients who received 150 mg or 300 mg of barzolvolimab for 52 weeks. Placebo comparison was done for the week 12 analysis.
- Statistical analyses to determine significance between groups was conducted using one-way ANOVA with Tukey's Multiple Comparisons Test, Welch's t-test, or z-test for two population proportions, as appropriate.
- Analyses shown reflect mean $\pm$ SE values.

Baseline characteristics median (IQR)	IgE normal/high (≥ 40 IU/mL)			IgE low (< 40 IU/mL)		
	Placebo (N=34)	150 mg Q4W (N=29)	300 mg Q8W (N=35)	Placebo (N=16)	150 mg Q4W (N=20)	300 mg Q8W (N=15)
Age, years	42.0 (33.3 - 50.3)	44.0 (37.0 – 48.0)	47.0 (37.5 - 57)	45.0 (31.5 - 60.8)	51.0 (42.8 - 62.3)	51.0 (46.5 - 55)
Female, n (%)	28.0 (82.3)	22.0 (75.9)	28.0 (80.0)	7.0 (43.8)	14.0 (70.0)	12.0 (80.0)
BMI	28.8 (24.9 - 35.2)	27.9 (24.6 – 33.0)	28.8 (26.0 – 34.0)	30.2 (25.2 - 34.7)	27.4 (24.1 - 29.8)	33.3 (27.2 - 35.5)
Tryptase (ng/mL)	4.4 (3.2 - 6.6)	5.7 (4.4 - 6.8)	4.7 (3.8 - 6.4)	4.8 (4.0 - 5.9)	5.6 (4.4 - 7.5)	5.7 (4.4 - 7.4)
Total IgE (IU/mL)	176.0 (99.8 - 317.3)	115.0 (66.0 – 222.0)	150.0 (87.5 – 258.0)	12.0 (3.5 - 20.8)	11.5 (5.5 - 18.3)	20.0 (3.5 – 35.0)
UAS7	32.0 (27.3 - 38.3)	30.0 (25.0 - 36.0)	30.3 (27.4 – 36.0)	26.0 (19.4 - 30.4)	33.0 (29.5 - 38.5)	31.0 (28.0 – 37.0)
UAS7, Severe Disease n (%)	25.0 (73.5)	19.0 (65.5)	26.0 (74.3)	7.0 (43.8)	16.0 (80.0)	13.0 (86.7)
UCT7	5.0 (4.0 – 6.0)	4.0 (1.5 - 6.5)	4.0 (1.0 -6.0)	6.0 (5.0 – 8.0)	5.5 (3.3 - 6.8)	3.0 (0 – 5.0)
Previous experience with omalizumab, n (%)	7.0 (20.6)	6.0 (20.7)	6.0 (17.1)	1.0 (6.0)	4.0 (21.0)	5.0 (33.3)

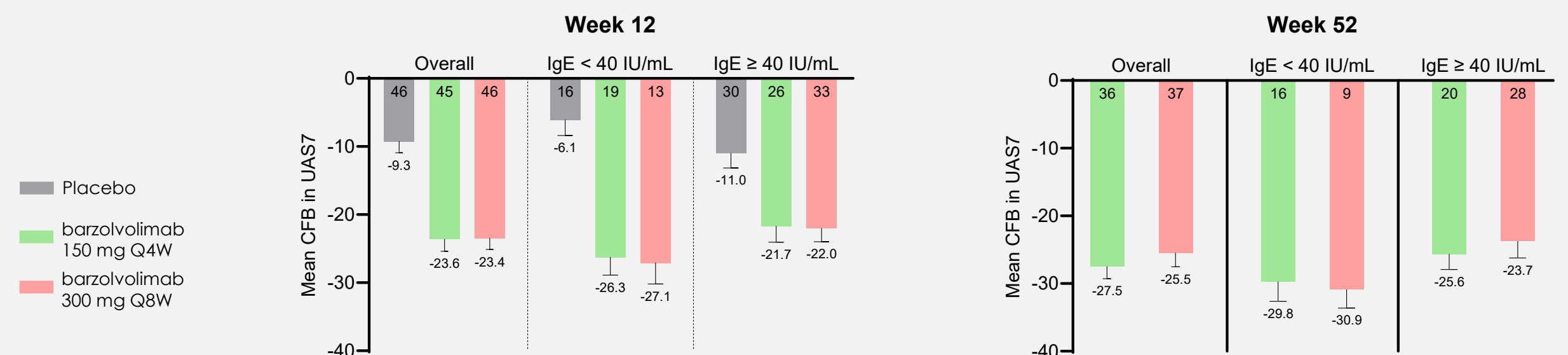
Results

Barzolvolimab exhibits similar efficacy in patients with normal/high vs. low total IgE levels**^

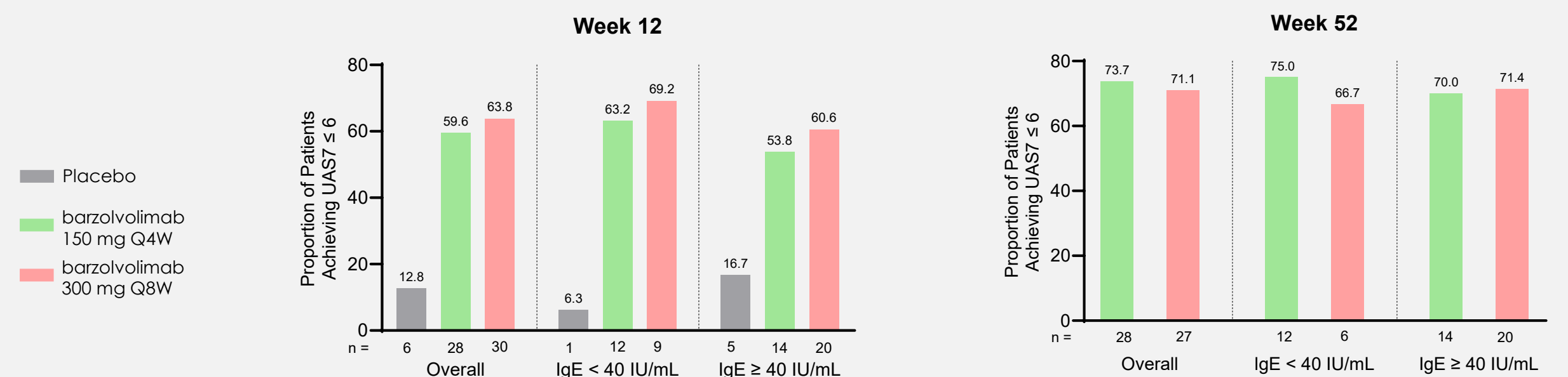
UAS7 Kinetics



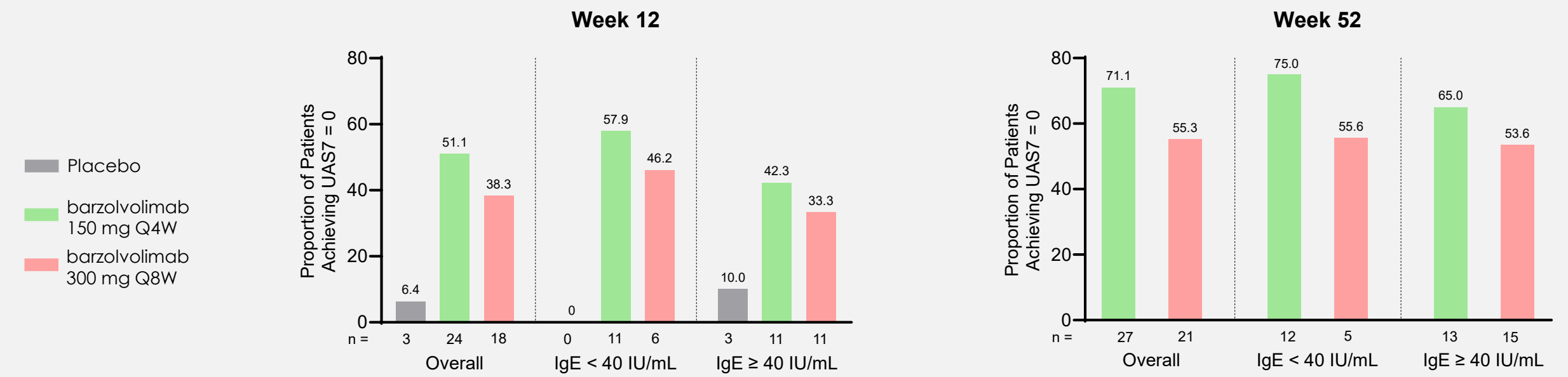
Mean change from baseline in UAS7



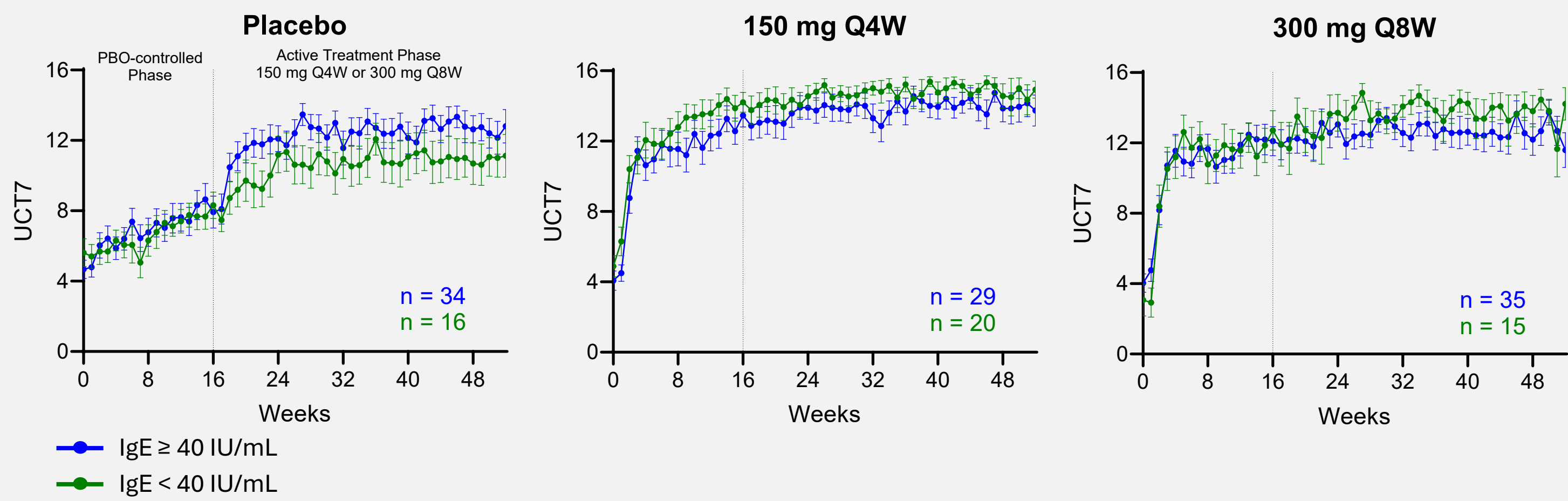
Proportion of patients achieving well-controlled disease (UAS7 ≤ 6)



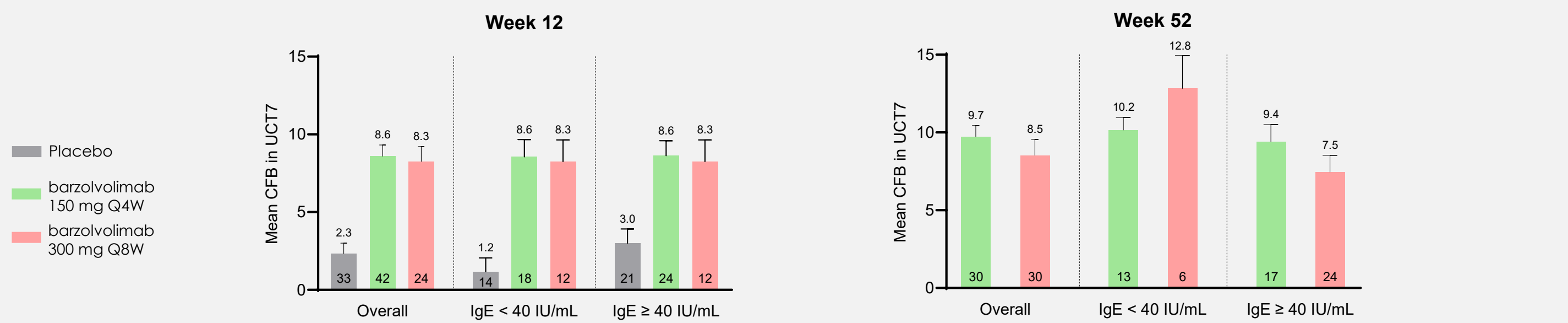
Proportion of patients achieving complete disease control (UAS7 = 0)



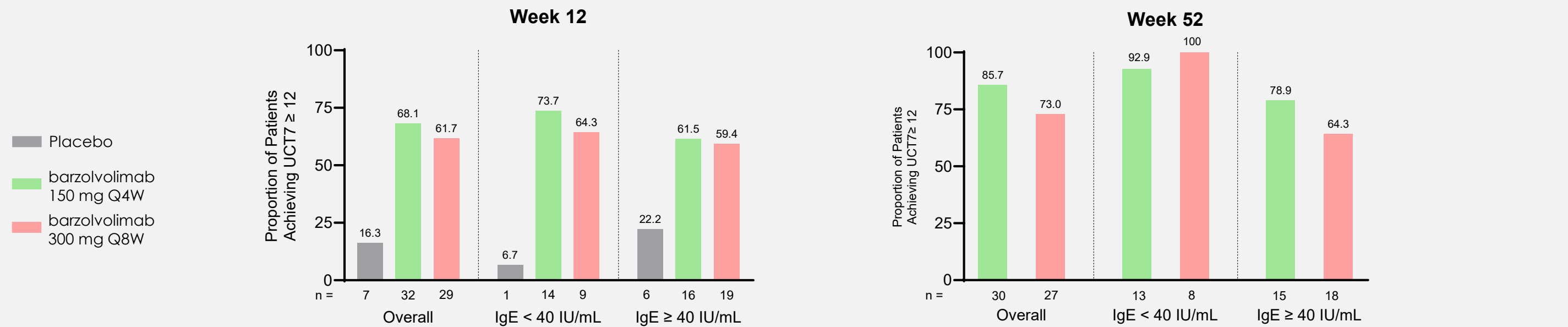
UCT7 Kinetics



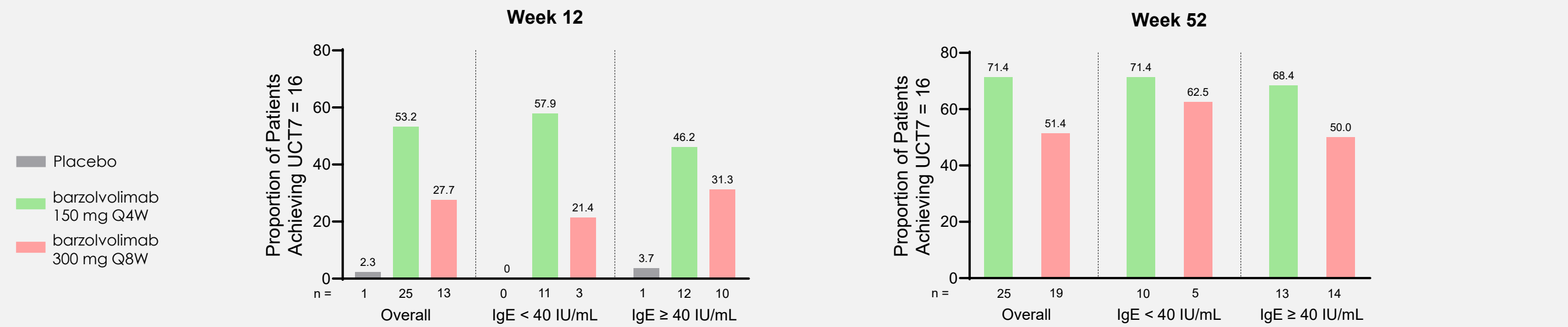
Mean change from baseline in UCT7



Proportion of patients achieving well-controlled disease (UCT7 ≥ 12)



Proportion of patients achieving complete disease control (UCT7 = 16)



* All comparisons between either dosing regimen vs. placebo in indicated analyses had an adjusted p-value < 0.01 , except for UAS7 = 0 between 300 mg vs placebo (p-value = 0.026) and UCT = 16 between 300 mg and placebo (p-value = NS).
+ Adjusted p-value was non-significant for all comparisons between normal/high vs IgE low subgroups in indicated analyses.
^ The rate ratio (95% CI) of achieving a complete response in patients with low compared to normal/high baseline IgE was 1.37 (0.74-2.55) and 1.24 (0.71-2.16) in the 150 mg Q4W and 300 mg Q8W groups, respectively.

Conclusions

- Barzolvolimab treatment demonstrated rapid and strong efficacy in a Phase 2 study in CSU and was well tolerated.
- Similarly strong efficacy was observed in patients with normal/high (≥ 40 IU/mL) or low (< 40 IU/mL) IgE levels for both dosing regimens as determined by:
 - Changes in UAS7 or UCT7 at Weeks 12 and 52
 - Rates of well-controlled or complete disease control
- These data suggest that mast cells are important drivers of CSU regardless of the underlying endotype and are consistent with the mechanism of action of barzolvolimab.
- Barzolvolimab is a promising therapy for all patients with moderate to severe CSU, regardless of underlying endotype.